📘 jQuery Selectors — Complete Instructor-Level Explanation

Since you're teaching **JavaScript + DOM**, understanding jQuery selectors deeply will help students connect CSS selectors with DOM manipulation.

1️⃣ What is a jQuery Selector?

A **jQuery selector** is used to select and manipulate HTML elements.

It uses **CSS selector syntax**, but returns a **jQuery object**.

Basic Syntax

$(selector)

- $ → jQuery function
- selector → CSS-style selector
- Returns → jQuery wrapped set (collection of matched elements)

2️⃣ Why jQuery Selectors Are Powerful

- Shorter than document.querySelectorAll
- Cross-browser consistent
- Supports advanced filtering
- Allows chaining
- Works with jQuery methods directly

3️⃣ Types of jQuery Selectors

We will classify them professionally:

1. Basic Selectors
2. Hierarchy Selectors
3. Attribute Selectors
4. Filter Selectors

5. Form Selectors

4️⃣ Basic Selectors

🔹 1. ID Selector

Selects element by id.

$("#header")

Example

<h1 id="title">Hello</h1>

$("#title").css("color","blue");

⚠ ID must be unique.

🔹 2. Class Selector

$(".box")

Example:

A

B

$(".box").hide();

All matching elements selected.

🔹 3. Element Selector

$("p")

Selects all <p> elements.

🔹 4. Universal Selector

$("*")

Selects all elements in the document.

⚠ Not recommended for performance-sensitive apps.

5️⃣ Hierarchy Selectors

These select elements based on relationships.

🔹 1. Descendant Selector

$("div p")

Selects all <p> inside .

🔹 2. Child Selector

$("div > p")

Selects only direct children.

🔹 3. Adjacent Sibling

$("h1 + p")

Selects first <p> immediately after <h1>.

🔹 4. General Sibling

$("h1 ~ p")

Selects all <p> after <h1>.

6️⃣ Attribute Selectors

Very powerful in real projects.

🔹 Select by Attribute

```
$("input[type='text']")
```

🔹 Attribute Equals

```
$("a[target='_blank']")
```

🔹 Attribute Starts With

```
$("img[src^='images/']")
```

🔹 Attribute Ends With

```
$("img[src$='.png']")
```

🔹 Attribute Contains

```
$("div[class*='card']")
```

7️⃣ Filter Selectors

Used to filter matched elements.

🔹 :first

```
$("p:first")
```

🔹 :last

```
$("p:last")
```

- **:even and :odd**

$("tr:even")

$("tr:odd")

Note: Index starts from 0.

- **:eq(index)**

$("li:eq(2)")

Selects 3rd element.

- **:gt() and :lt()**

$("li:gt(2)") // greater than index 2

$("li:lt(3)") // less than index 3

---

8 Form Selectors

Extremely important in real applications.

- **:input**

$(":input")

Selects all form elements.

- **:text**

$(":text")

- **:password**

$(":password")

🔹 **:checked**

```
$(":checked")
```

Useful for checkboxes & radio buttons.

🔹 **:selected**

```
$("option:selected")
```

9️⃣ Combining Selectors

You can combine selectors for precision.

```
$("div.box:first")
$("ul li.active")
$("input[type='text']:first")
```

🔟 Real-Time Practical Demo

HTML

```
<ul id="menu">
  <li class="item">Home</li>
  <li class="item">About</li>
  <li class="item active">Contact</li>
</ul>
```

jQuery

```
$("#menu li.active").css("color","red");
```

Explanation:

- Select ul with id menu
- Inside it select li

- Filter class active

🔎 jQuery vs Modern JavaScript

jQuery	Modern JS
$("#id")	document.getElementById()
$(".class")	document.querySelectorAll()
$("div p")	document.querySelectorAll()